

LOCAL BUS MANAGEMENT SYSTEM

Objectives

- **Efficient Fleet Management:**
Keeps track of buses, their routes, schedules, and maintenance status.
- **Better Customer Experience:**
Provides real-time bus tracking
- **Reduces paperwork for employee Records:**
Maintains information on drivers, conductors, and other staff.
- **Data Management & Security:**
Stores data securely for better decision-making.
- **Easy communication and feedback of passengers:**
Provides easy communication to passengers and easily share feedback

Input of the project

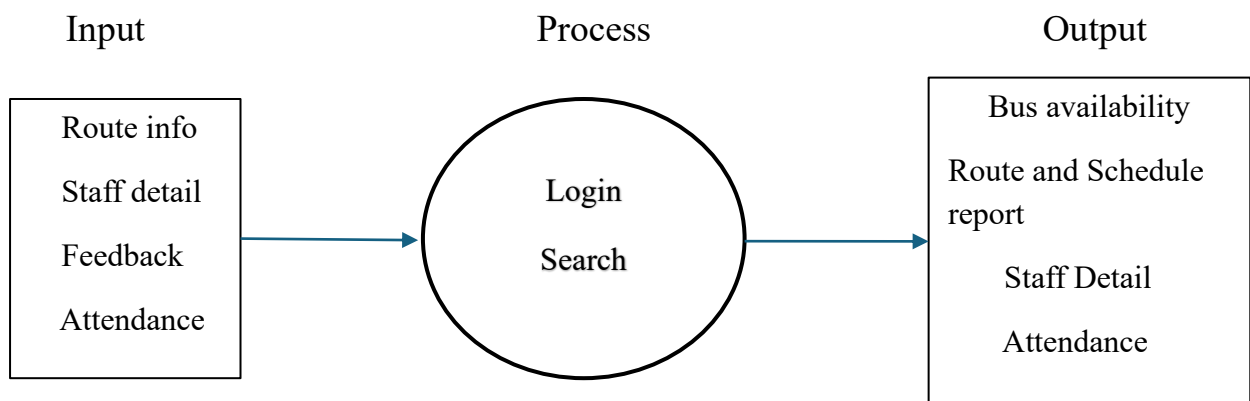
- **Route Information:**
Start and end locations.
- **Staff Details:**
Name, ID, contact information, job role.
- **Comments and feedbacks.**
- **Attendance of the staff.**

Output of the project

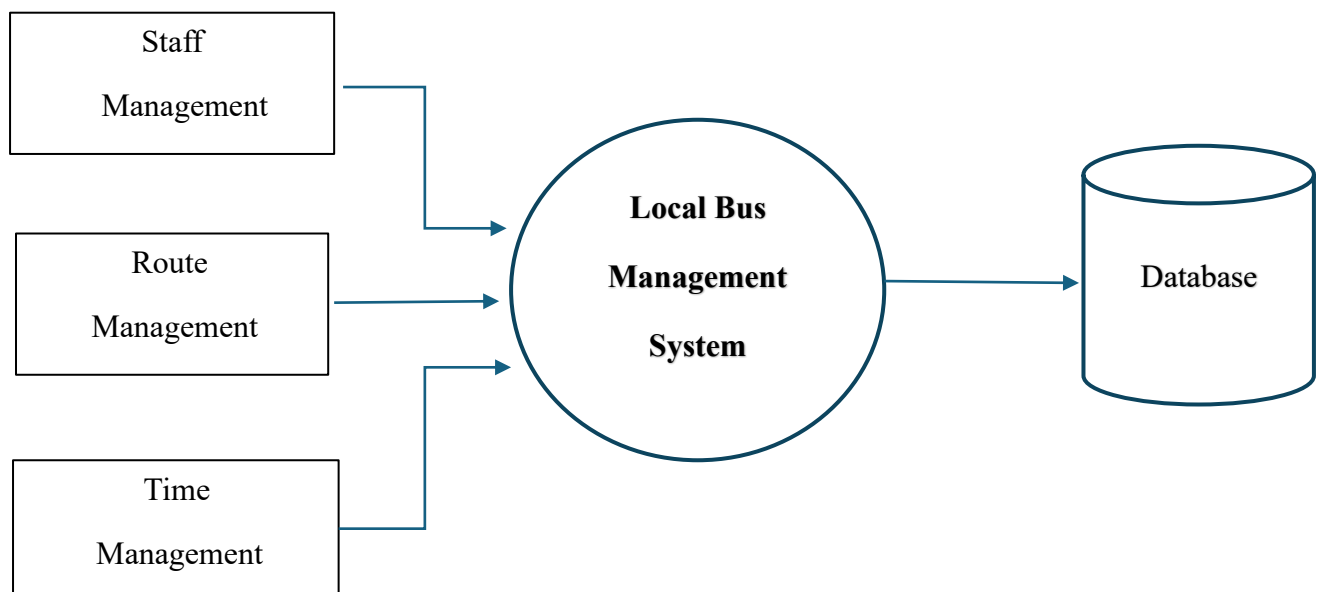
- **Bus Availability Status**
- **Route and schedule report**

- Staff details
- Attendance of the staff

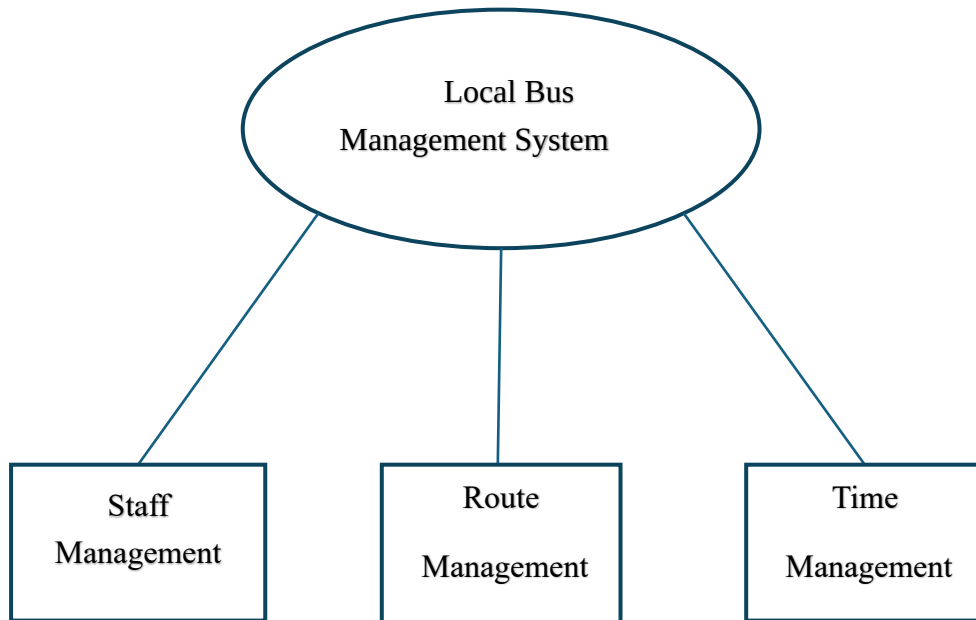
Process Logic



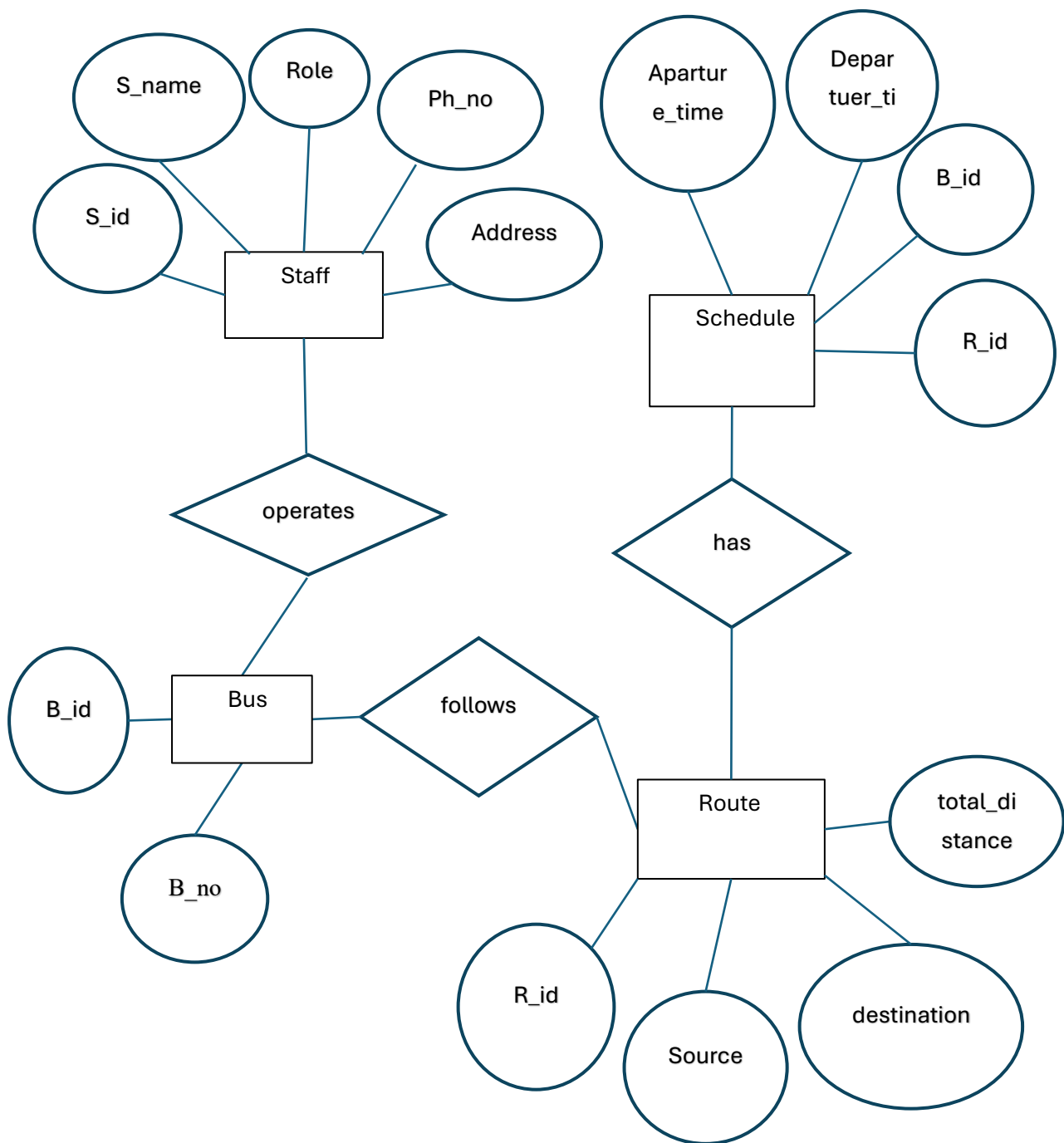
Data Flow Diagram (0 Level)



Context model



ER Diagram



Required tools and platforms

Front end

- CSS
- JavaScript
- HTML
- React

Back end

- MySQL
- PHP
- Node.js
- Python
- Mongo DB

IDEs and software

- Thonny
- WampServer
- Notepad
- V S code
- PyCharm

Hardware Requirements

- ◇ Processor: Dual-core processor
- ◇ RAM:2gb or more
- ◇ Storage:512gb
- ◇ System type: Windows 7 or latest version

Are you doing this project for a client /organization?

YES,

This project, Local Bus Management System is being developed for an organization named “NWKRTC”. Based on the needs and feedback provided by the organization, ensuring that it aligns with their workflow and enhances their management capabilities. The system will be user-friendly, secure, and accessible, catering to both administrator and members.

Duration of the Project

☐ 2 months

Members of the project

☐ 1 Member

Limitation of the project

- ◆ Limited Real-Time Information
- ◆ The system may not always provide real-time information, which can lead to delays or inaccuracies.
- ◆ Operational Limitations:
Limited flexibility in responding to changes or disruptions

Scope of the Project

- ◆ Include GPS tracking devices, mobile devices, and servers.
- ◆ Online seat reservation and payments.

